



Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology

Course Code & Title : IT3501 Full Stack Web Development

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 2 / Implementing HTTP Services in Node.js
- **Course Outcome** : CO2
- **Topic Learning Outcome:** TLO9
- **Activity Chosen** : One Minute Paper
- **Justification:**
 - The one-minute paper is a highly effective technique for fostering active learning and quick reflection.
 - For a topic like HTTP services in Node.js, this activity encourages students to consolidate their knowledge by capturing the essence of how HTTP requests and responses work in a real-time programming environment.
 - In order to have a recap of HTTP services in Node.js, this activity is chosen.
- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
 - Faculty explained the concept of how HTTP services are created and managed in Node.js, offering flexibility and performance for modern web development.
 - Based on the discussion and after clarifying the student's doubts, the faculty asked the students to write the concept and simple program using node.js today as shown in Figure 1: Node.js program to demonstrate the implementation of HTTP services.
 - The faculty member collected the paper and corrected the programs as shown in Fig 2. The students who have written program with mistakes were asked to rewrite it

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	2	3	2	1	3

(1 – Low 2 – Moderate 3 – High)



• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students will be able to understand the concept of HTTP services and apply to real time programming environments	Students will be able to write and develop simple node.js program using http module	Students will be able to develop server-side application programs	As the student involved in the activity as an individual, the individual programming skill improved	Students will be able to communicate effectively on complex engineering activities	The node.js programming skill gained through this activity helps the students in solving real world problems

• Images / Screenshot of the practice:

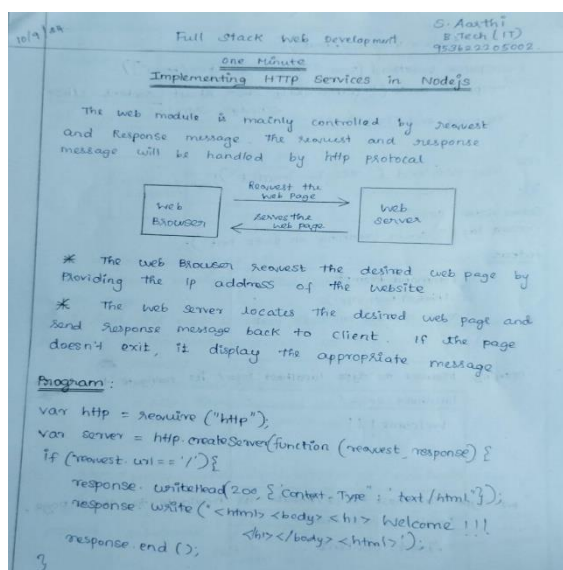


Fig 1: One Minute Paper Activity



Fig 2: Sample Sheet of Activity of Student Aarthi S

• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Students said that the activity assisted them in determining their level of understanding of the concept.
- They said it would be useful to recall the concept in short time.
- Students feel easy to understand the concept and to write node.js programs.



❖ ***Benefit of the practice:***

- This activity enables students to understand and summarize key ideas such as how Node.js handles incoming HTTP requests and provide responses.
- The one-minute paper activity made the slow learners to remember the concept easily.
- From this activity, Students learned how to implement HTTP services using Node.js

❖ ***Challenges faced in implementation:***

- Few students made a mistake in writing HTTP services programs using node.js
- It took more time for some students to complete the activity.

The Student Thirumurugan.A answered the question implementation of HTTP services in Node.js with following programming mistakes

- i) He committed a mistake in creating simple server in Node.js
- ii) He committed a mistake in requiring node.js modules.

In the next class, I addressed the students how to create a simple server in node.js using http built-in modules.

References:

- https://www.tutorialspoint.com/the_full_stack_web_development/index.asp
- <https://www.udemy.com/course/the-full-stack-web-development/>

Signature of Faculty Member

HOD